

## Python for Data Science (PDS)

### Lab - 14

Roll No. : 128

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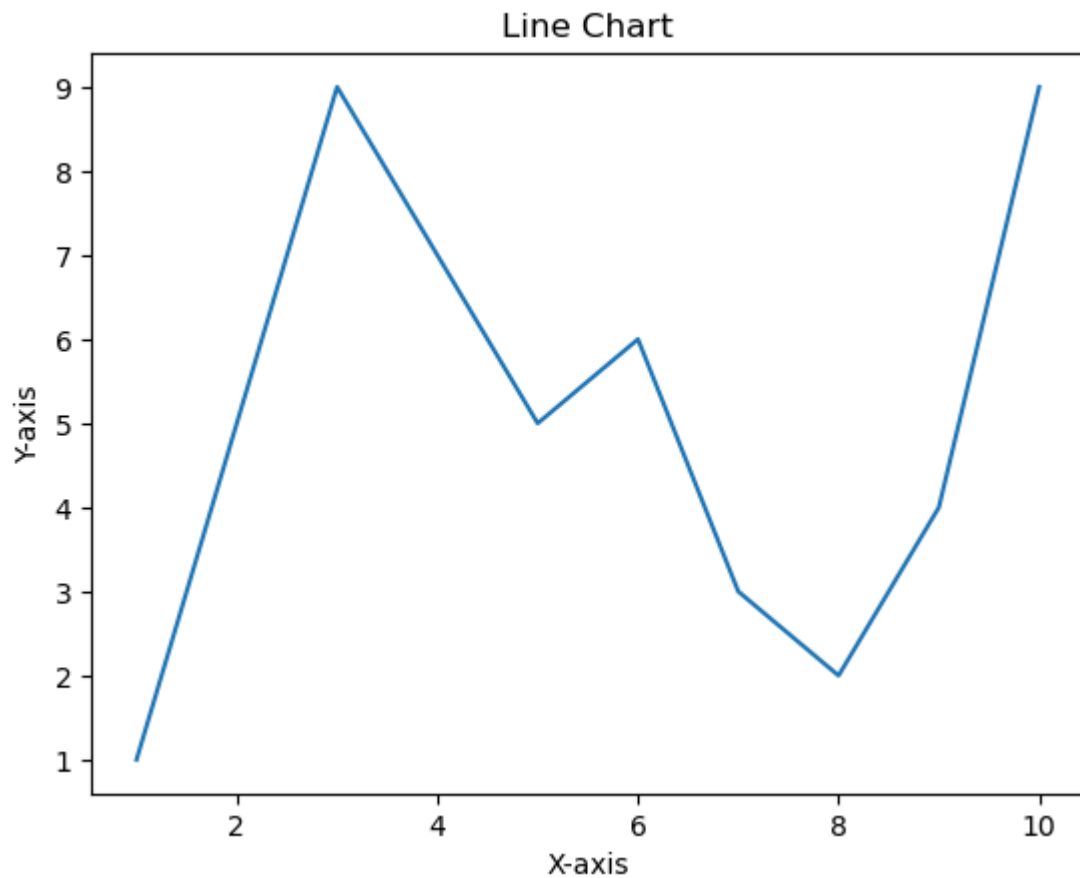
### Import matplotlib below

```
In [12]: import matplotlib.pyplot as plt  
import numpy as np
```

01) WAP to display the line chart for given data for x and y axis.

```
In [13]: x = range(1,11)  
y = [1,5,9,7,5,6,3,2,4,9]
```

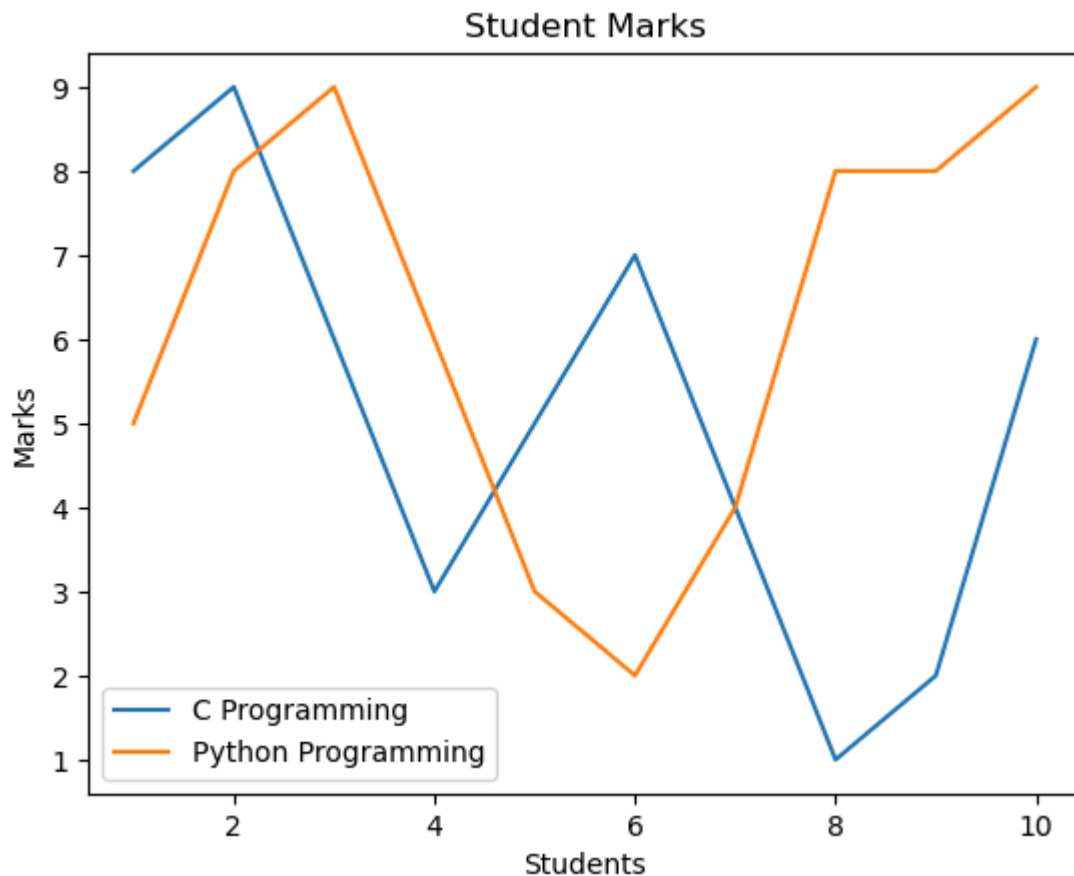
```
In [14]: plt.plot(x, y)  
plt.xlabel("X-axis")  
plt.ylabel("Y-axis")  
plt.title("Line Chart")  
plt.show()
```



02) WAP to generate line chart for given data.

```
In [15]: x = range(1,11,1)
cxMarks= [8,9,6,3,5,7,4,1,2,6]
cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

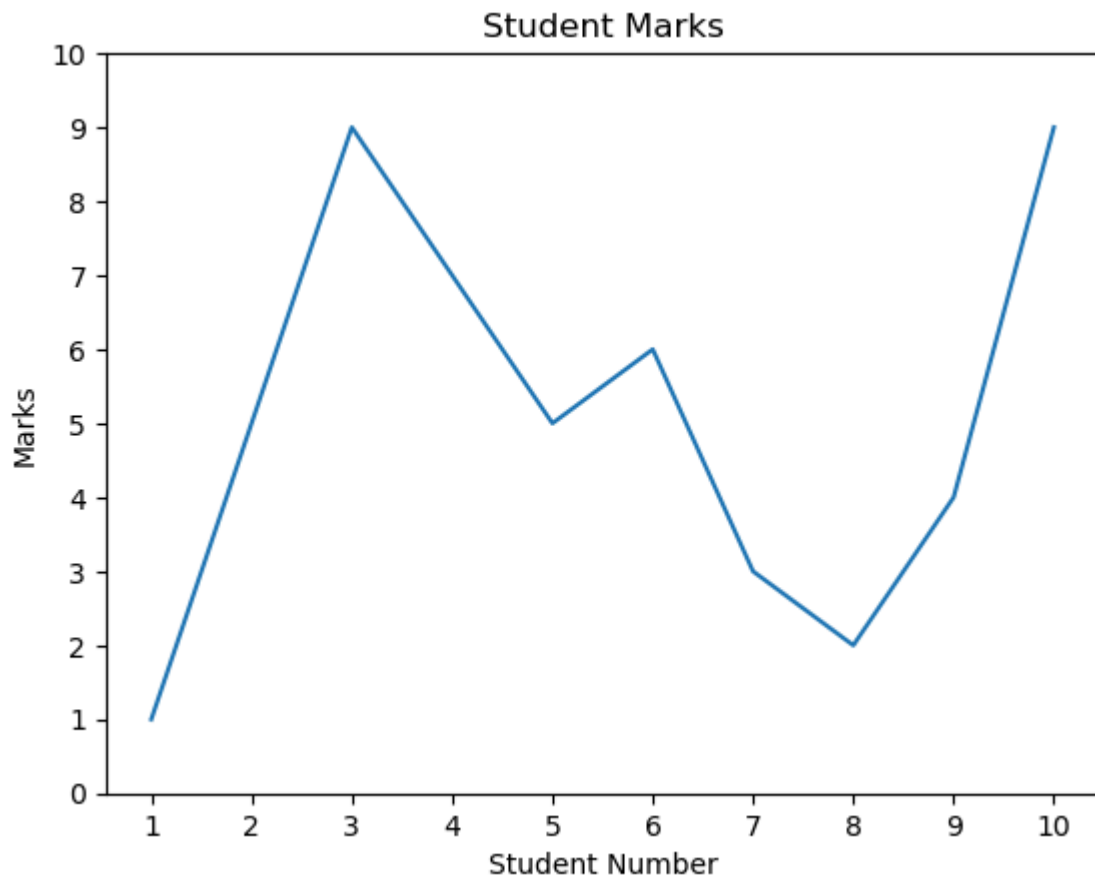
```
In [18]: plt.plot(x, cxMarks, label="C Programming")
plt.plot(x, cyMarks, label="Python Programming")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.legend()
plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

```
In [16]: x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

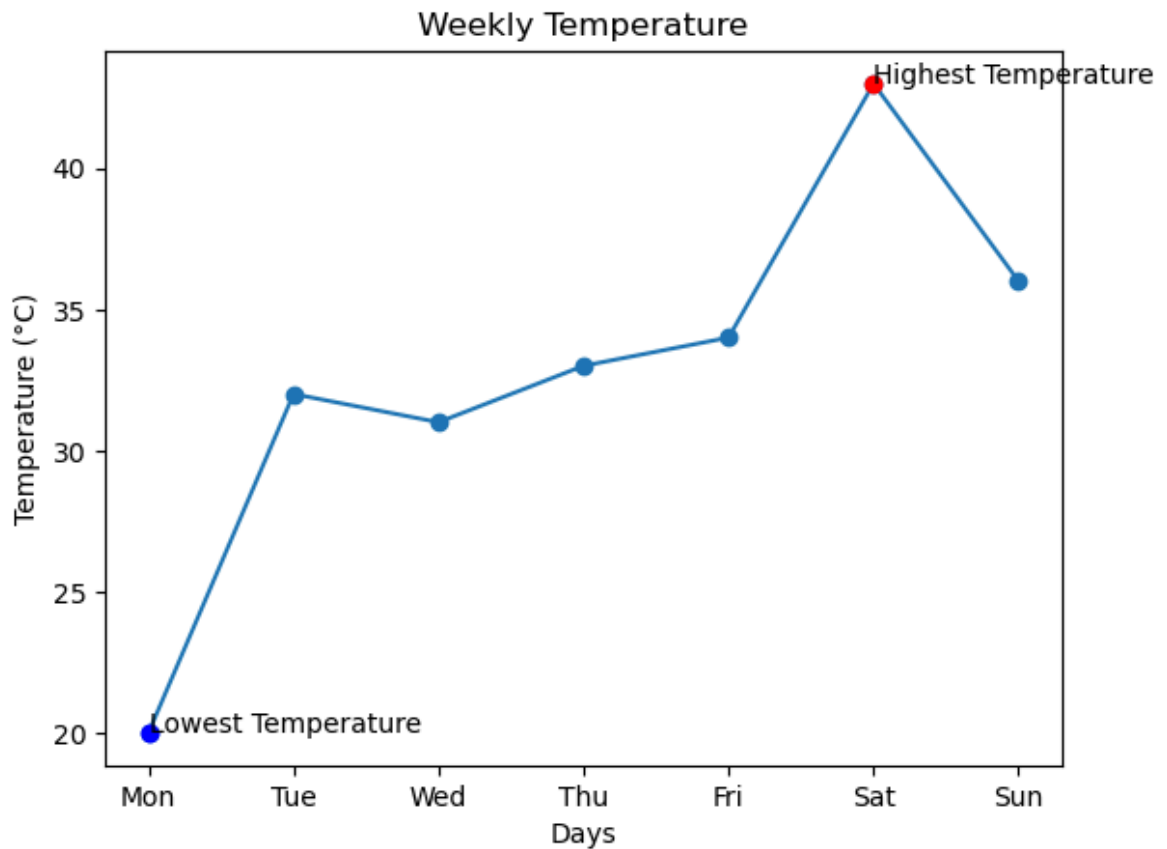
```
In [19]: plt.plot(x, y)
plt.xticks(range(1, 11, 1))
plt.yticks(range(0, 11, 1))
plt.xlabel("Student Number")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```



**04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperature.**

```
In [20]: Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
         Temperature = [20, 32, 31, 33, 34, 43, 36]
```

```
In [21]: plt.plot(Days, Temperature, marker="o")
         plt.plot("Mon", 20, marker="o", color="blue")
         plt.plot("Sat", 43, marker="o", color="red")
         plt.annotate("Lowest Temperature", xy=("Mon", 20))
         plt.annotate("Highest Temperature", xy=("Sat", 43))
         plt.xlabel("Days")
         plt.ylabel("Temperature (°C)")
         plt.title("Weekly Temperature")
         plt.show()
```

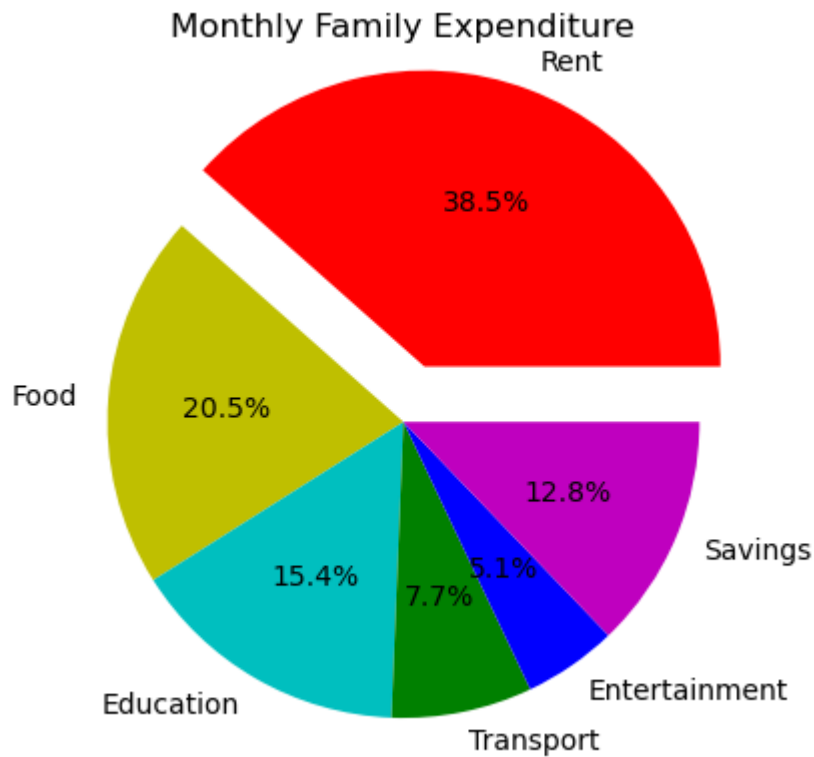


05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

```
In [23]: Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Saving"]  
Amount = [15000, 8000, 6000, 3000, 2000, 5000]
```

```
In [27]: plt.pie(Amount,  
               labels=Categories,  
               colors=["r", "y", "c", "g", "b", "m"],  
               explode=[0.2, 0, 0, 0, 0, 0],  
               autopct="%1.1f%%",  
               radius=1)  
plt.title("Monthly Family Expenditure")  
plt.show()
```



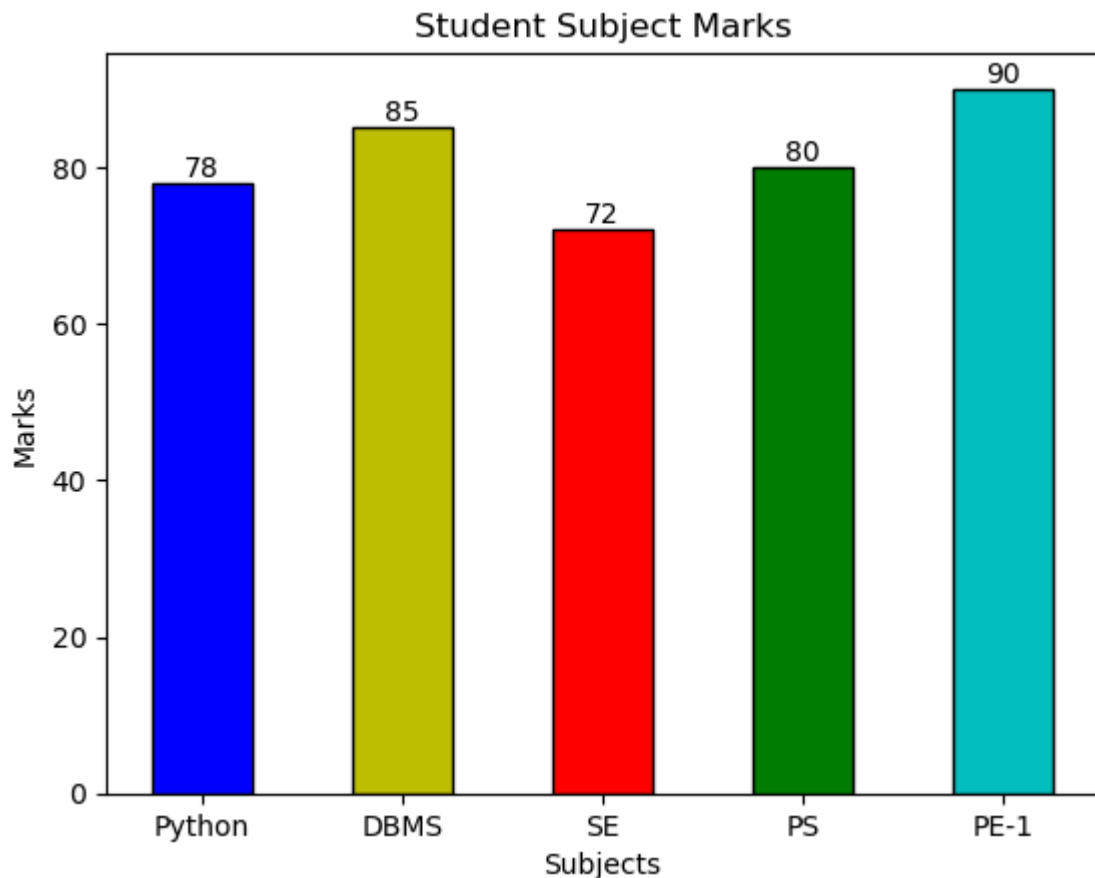
06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

```
In [31]: Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
Marks = [78, 85, 72, 80, 90]
```

```
In [33]: bar = plt.bar(
    Subjects,
    Marks,
    width=0.5,
    color=["b", "y", "r", "g", "c"],
    edgecolor="black"
)

plt.bar_label(bar)
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("Student Subject Marks")
plt.show()
```

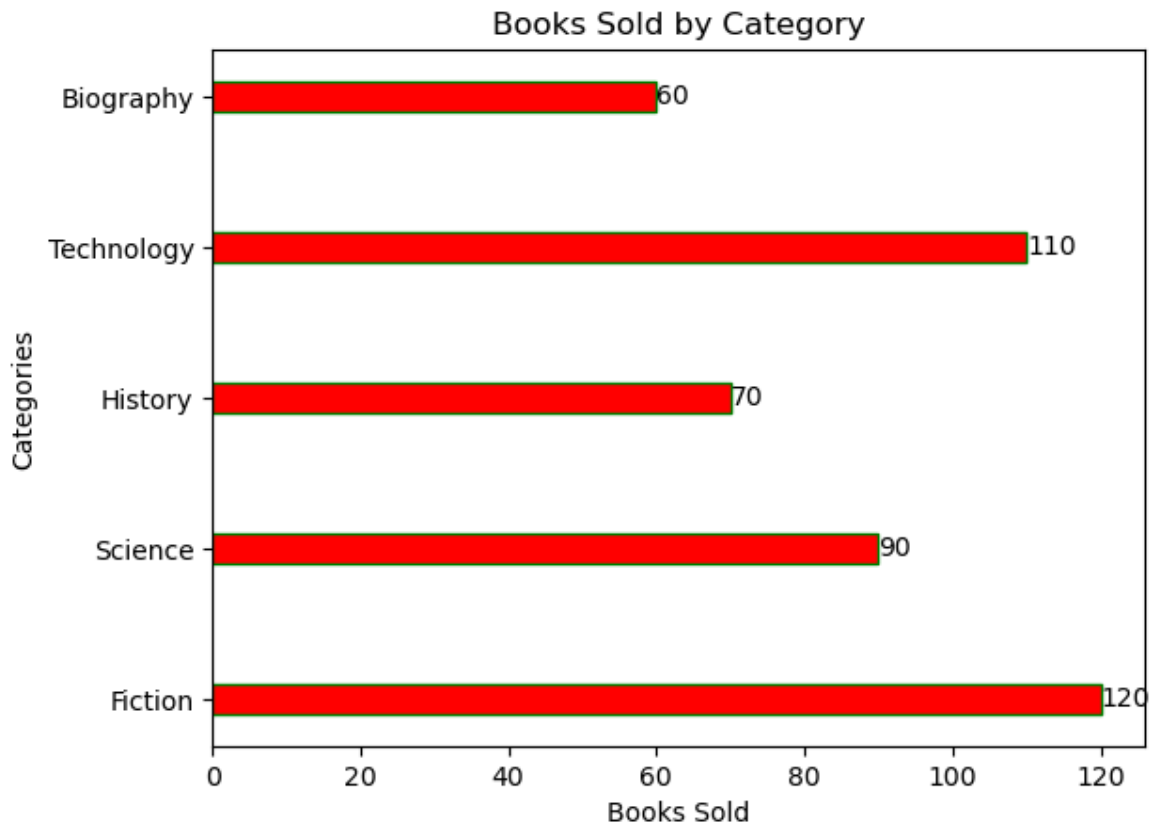


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

```
In [36]: Categories = ["Fiction", "Science", "History", "Technology", "Biography"]  
Books_Sold = [120, 90, 70, 110, 60]
```

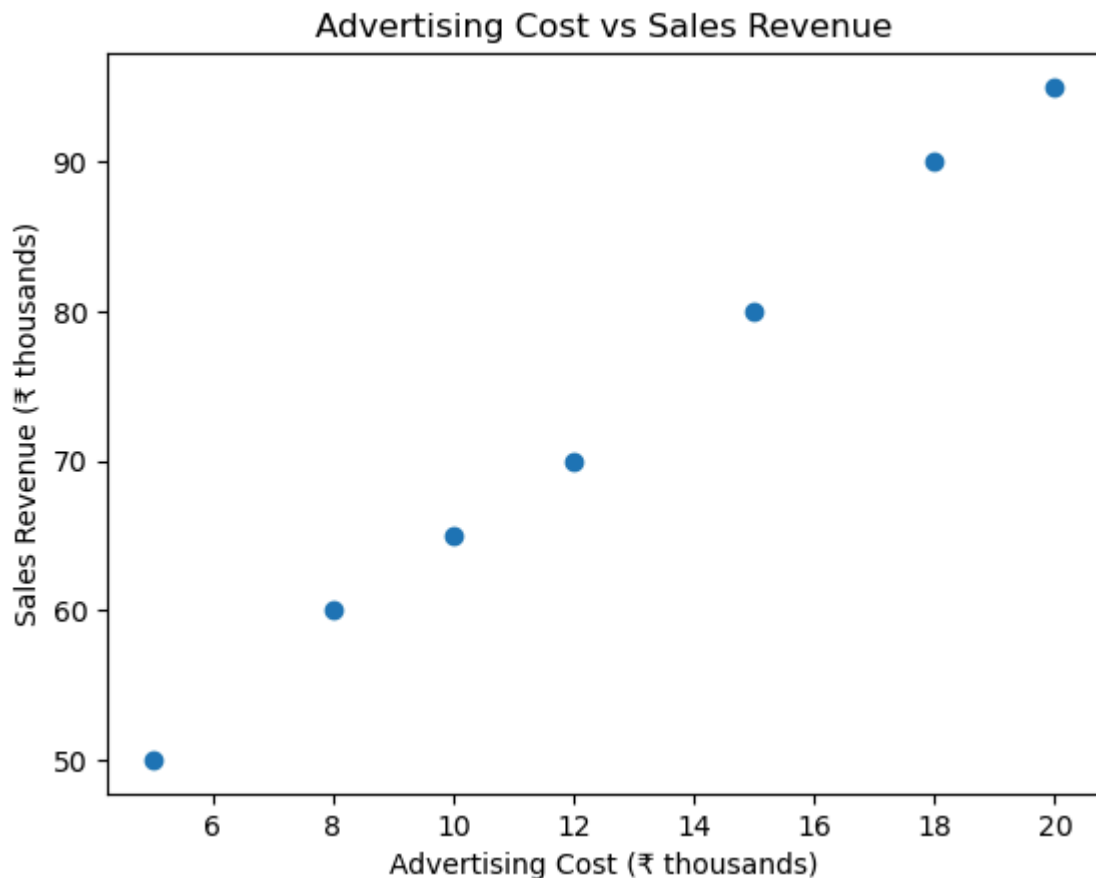
```
In [39]: bar = plt.barh(Categories, Books_Sold, color = "r", edgecolor = "g", height=0.2)  
plt.bar_label(bar)  
plt.xlabel("Books Sold")  
plt.ylabel("Categories")  
plt.title("Books Sold by Category")  
plt.show()
```



**08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.**

```
In [44]: Advertising_Cost = [5, 8, 10, 12, 15, 18, 20]
Sales_Revenue = [50, 60, 65, 70, 80, 90, 95]
```

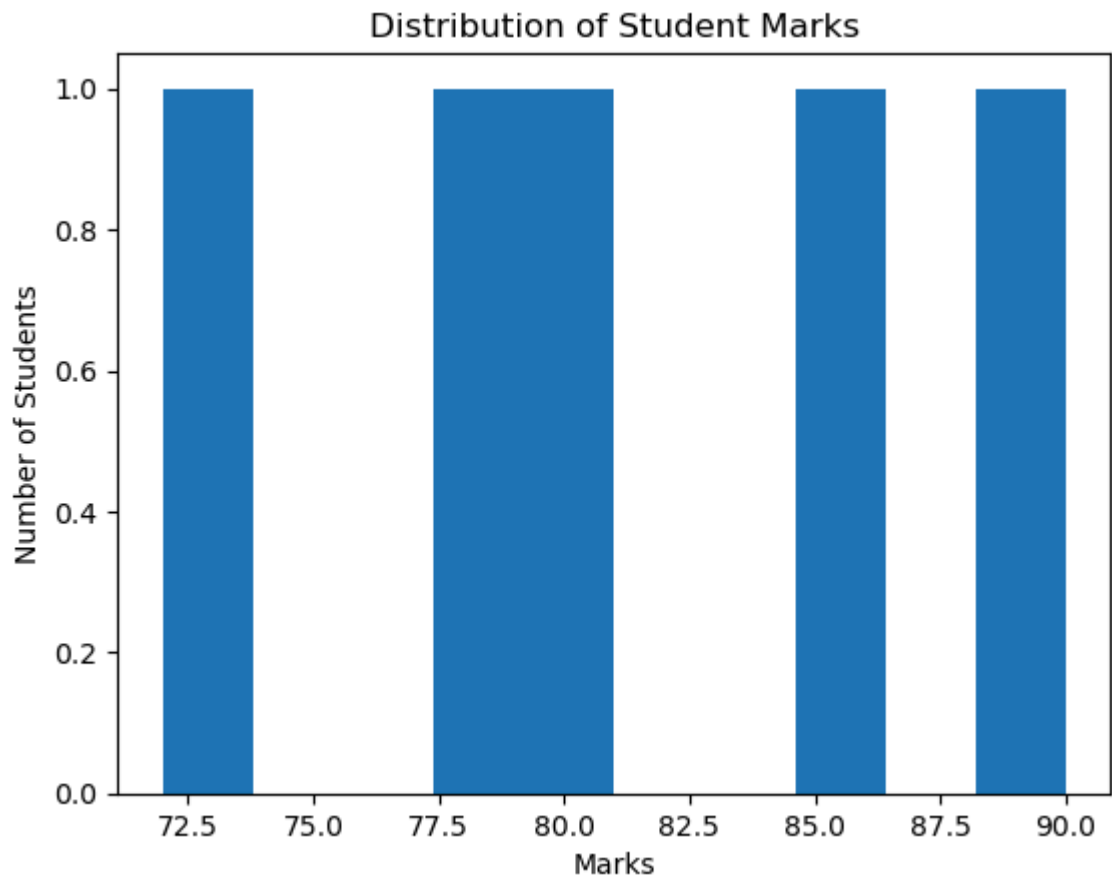
```
In [45]: plt.scatter(Advertising_Cost, Sales_Revenue)
plt.xlabel("Advertising Cost (₹ thousands)")
plt.ylabel("Sales Revenue (₹ thousands)")
plt.title("Advertising Cost vs Sales Revenue")
plt.show()
```



**09) WAP to create a histogram representing the distribution of marks obtained by students in a class.**

```
In [ ]: Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

```
In [46]: plt.hist(Marks)
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Student Marks")
plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

```
In [49]: Commuting_Time = [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 90]
```

```
In [50]: plt.boxplot(Commuting_Time)
plt.ylabel("Commuting Time (minutes)")
plt.title("Daily Commuting Time")
plt.show()
```

Daily Commuting Time

